Affine Constraint Dynamics and Density-One Descent in the Collatz Map

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April 22, 2026

Abstract

We study the Collatz map via the fully accelerated transformation

$$f(x) = \frac{3x+1}{2^{v_2(3x+1)}}.$$

Finite trajectories are encoded by division words, which record the 2-adic valuations encountered along an orbit. Each such word induces an affine map whose admissibility imposes a congruence constraint on initial values. This yields a symbolic–affine framework in which admissible words correspond to residue classes modulo powers of two.

We show that these constraints refine exponentially: a division word of length m restricts initial values to a residue class modulo a power of 2 that is comparable to, but strictly larger than, 3^m, while the number of admissible words grows like ρ^m with $\rho < 3$.

This creates a competition between combinatorial growth and arithmetic refinement: while the number of admissible words grows like ρ^m with $\rho < 3$, each word imposes a congruence constraint modulo a power of 2 that grows strictly faster than 3^m. This imbalance forces exponential sparsification of admissible initial conditions.

As a consequence, the set of integers that fail to admit a finite descent prefix has natural density zero. In particular, almost all integers eventually decrease under iteration of the Collatz map.

Contents

1 Introduction

The $3x + 1$ problem, also known as the Collatz conjecture [1], concerns the behavior of the map

$$T(x) = \begin{cases} \dfrac{x}{2} & \text{if } x \text{ is even,} \\ 3x + 1, & \text{if } x \text{ is odd.} \end{cases}$$

on the positive integers. It asserts that every positive integer eventually reaches 1 under iteration of T. Despite its elementary formulation, the conjecture has resisted proof for decades [2, 3]. Extensive computational verification confirms convergence for very large ranges [4], but does not reveal the global structure governing trajectory behavior.

A major advance by Tao [5] showed that almost all integers, in the sense of logarithmic density, eventually decrease under iteration of the Collatz map. This result is obtained via probabilistic averaging rather than an explicit structural decomposition of the integers.

The present work develops a deterministic, arithmetic framework for analyzing typical Collatz behavior based on a symbolic–affine encoding of trajectories. Using this approach, we encode the dynamics symbolically rather than studying individual trajectories with the fully accelerated map

$$f(x) = \frac{3x + 1}{2^{v_2(3x+1)}},$$

which maps odd integers to odd integers by compressing each odd step into a single transformation.

Each trajectory under f is represented by a finite *division word*, recording the number of factors of 2 removed (the 2-adic valuation) at each iterate.

These words admit an exact affine representation obtained by symbolic composition of the accelerated map: each division word induces a map of the form

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}},$$

and determines a corresponding congruence class modulo $2^{D(\omega)}$. Thus division words determine canonical arithmetic *constraint classes* (residue classes) in the integers, once realizability is imposed.

This correspondence allows the dynamics to be studied at the level of residue classes rather than individual orbits. Convergent division words—those for which the associated affine map is contractive—identify families of integers that decrease after a fixed number of steps. The global problem is thereby reduced to understanding how these families are distributed and how their densities accumulate.

Two competing effects govern this distribution. On one hand, the number of admissible convergent division words grows exponentially with length. On the other hand, each word imposes a congruence constraint whose modulus grows exponentially with the total 2-adic valuation. Using structural constraints on admissible division words, encoded by an increment grammar, we show that the combinatorial growth rate is strictly subcritical: the number of admissible words grows like ρ^m for some $\rho < 3$, while the moduli scale like powers of 2 satisfying $3^m < 2^{D(\omega)} < 2 \cdot 3^m$.

This transforms the Collatz problem into a competition between symbolic growth and arithmetic refinement, with convergence governed by their relative exponential rates.

This imbalance leads to *exponential sparsification*: the proportion of integers admitting a convergent division word of length m decays exponentially in m. Summing over all lengths then shows that the complement—integers that avoid all such descent patterns—has natural density zero.

Consequently, the set of integers whose trajectory under iteration of f eventually decreases has natural density one.

While this does not exclude the existence of exceptional trajectories, any such exceptions must lie within a set of zero density and cannot arise from any scalable arithmetic structure generated by the symbolic dynamics.

Structure of the Paper. The argument proceeds as follows:

- Section 2 introduces division words and their total division counts.
- Section 3 develops the affine representation of division words and their congruence structure.
- Section 4 establishes admissibility and minimal valuation constraints.
- Section 5 controls the growth of admissible words via the increment grammar.
- Section 6 identifies the residue class structure induced by division words.

- Section 7 proves exponential sparsification of admissible sets.
- Section 8 establishes density-one descent.

2 Symbolic Encoding of Collatz Trajectories

2.1 Division Words

Definition 2.1 (Division Word)**.** A division word of length $m \geq 0$ is a sequence

$$\omega = (d_0, \dots, d_{m-1}),$$

where each $d_i \in \mathbb{N}$ is intended to represent the number of factors of 2 removed at the i-th iterate of the accelerated map

$$f(x) = \frac{3x+1}{2^{v_2(3x+1)}}.$$

Remark. Since f maps odd integers to odd integers, each $3f^i(x)+1$ is even, so $d_i \geq 1$ for all i.

Definition 2.2 (Realization)**.** An odd integer $x \in \mathbb{N}$ realizes ω if

$$d_i = v_2(3f^i(x)+1), \quad 0 \leq i < m.$$

2.2 Total Division Count

Definition 2.3. The total division count of a word ω is

$$D(\omega) := \sum_{i=0}^{m-1} d_i.$$

Remark. The quantity $D(\omega)$ measures the total dyadic contraction associated with the symbolic path ω.

3 Affine Representation of Division Words

Definition 3.1. Each division word ω of length m induces an associated affine map

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}},$$

obtained by symbolic composition of the accelerated map.

Proposition 3.1. The map $\omega \mapsto F_\omega$ defines a semigroup homomorphism from concatenation of division words (read left to right) to composition of affine maps.

3.1 Example

Consider convergent segments with $m = 4$ iterates of f. The minimal total division count for $m = 4$ is

$$A020914(4) = 7.$$

One admissible division word attaining this minimum is

$$\omega = (1, 1, 1, 4),$$

for which

$$D(\omega) = 1 + 1 + 1 + 4 = 7.$$

Composing the accelerated map according to ω yields

$$f_\omega(x) = \frac{3^4 x + c(\omega)}{2^7} = \frac{81x + 65}{128},$$

so the canonical affine constant for this word is

$$c(\omega) = 65.$$

The integrality condition

$$81x + 65 \equiv 0 \pmod{128}$$

determines a unique residue class

$$x \equiv 15 \pmod{128}.$$

This congruence defines a constraint fiber with modulus 2^7.

Interpretation. The word $\omega = (1, 1, 1, 4)$ means:

- At each of the first three iterates of f, $3x + 1$ contains exactly one factor of 2.
- At the fourth iterate of f, $3x + 1$ contains four factors of 2.

Under the accelerated map

$$f(x) = \frac{3x+1}{2^{v_2(3x+1)}},$$

all of these divisions occur within a single iterate. Thus $d_3 = 4$ means that the fourth iterate divides by 2^4 in that step. The constant $c(\omega)$ is fixed entirely by this symbolic structure and ensures that precisely the integers in the residue class $x \equiv 15 \pmod{128}$ produce integer iterates under the affine form.

This example illustrates that the behavior of trajectories is governed entirely by the symbolic data encoded in ω and its associated affine map. The initial values that realize a given word are precisely those satisfying the induced congruence constraint, independent of any particular choice of representative.

In general, not every division word arises from an actual trajectory; the condition for realizability will be formulated later as an explicit congruence constraint, defining the class of *admissible* words.

4 Affine Symbolic Admissibility Grammar

Remark (Division Grammar as a Dynamical Encoding). Recall from Section 2 that a *division word* $\omega = (d_0, \ldots, d_{m-1})$ records the exact 2-adic valuation $d_i = v_2(3x_i + 1)$, occurring at each iterate of the fully accelerated Collatz map.

Unlike a purely combinatorial encoding, division words retain essential dynamical information. Each division word $\omega = (d_0, \ldots, d_{m-1})$ determines the sequence of multiplicative factors of the linear coefficient

$$r_i = \frac{3}{2^{d_i}},$$

so that the contractive or expansive contribution of each iterate is preserved in the linear coefficient (with the affine term handled separately via $c(\omega)$). This structure enables quantitative control of admissible trajectories via affine growth and to formulate sufficient conditions for descent, even though individual trajectories are collapsed under the encoding.

4.1 Explicit Form of the Affine Constant

Lemma 4.1 (Explicit Form of the Affine Constant)**.** *Let* $\omega = (d_0, \ldots, d_{m-1})$ *be a division word. Then the affine map*

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}}$$

has the explicit form

$$c(\omega) = \sum_{j=0}^{m-1} 3^{m-1-j} \cdot 2^{\sum_{i=0}^{j-1} d_i},$$

where empty sums are defined as 0*.*

4.2 Injectivity of the Affine Encoding

Lemma 4.2 (Affine Encoding Injectivity at Fixed Length)**.** *Let* $\omega = (d_0, \ldots, d_{m-1})$ *and* $\omega' = (d'_0, \ldots, d'_{m-1})$ *be division words of the same length* m*. Let*

$$c(\omega) = \sum_{j=0}^{m-1} 3^{m-1-j} \cdot 2^{D_j}, \quad D_j = \sum_{i=0}^{j-1} d_i,$$

and define $c(\omega')$ *analogously.*

If $\omega \neq \omega'$*, then*

$$c(\omega) \neq c(\omega').$$

Equivalently, the map $\omega \mapsto c(\omega)$ *is injective on admissible division words of fixed length* m*.*

Proof. Suppose $\omega \neq \omega'$, and let

$$k = \min\{\, j : d_j \neq d'_j \,\}$$

be the first index at which the two words differ. Then for all $j < k$, we have $D_j = D'_j$, while $D_k \neq D'_k$.

Consider the difference

$$\Delta := c(\omega) - c(\omega') = \sum_{j=0}^{m-1} 3^{m-1-j} \big(2^{D_j} - 2^{D'_j}\big).$$

The terms with $j < k$ cancel, so

$$\Delta = 3^{m-1-k}\big(2^{D_k} - 2^{D'_k}\big) + \sum_{j=k+1}^{m-1} 3^{m-1-j}\big(2^{D_j} - 2^{D'_j}\big).$$

Let $\delta_k = \min(D_k, D'_k)$. Without loss of generality, assume $D_k > D'_k$. Then

$$2^{D_k} - 2^{D'_k} = 2^{D'_k}(2^{D_k - D'_k} - 1),$$

where $2^{D_k - D'_k} - 1$ is odd. Hence

$$v_2(2^{D_k} - 2^{D'_k}) = D'_k = \delta_k.$$

For $j > k$, both D_j and D'_j strictly exceed δ_k, since the prefix sums increase by positive integers. Therefore

$$v_2\Big(3^{m-1-j}\big(2^{D_j} - 2^{D'_j}\big)\Big) > \delta_k \quad \text{for all } j > k.$$

It follows that Δ is a sum of one term with 2-adic valuation exactly δ_k and remaining terms of strictly higher 2-adic valuation, so no cancellation at the lowest 2-adic order is possible. Hence

$$v_2(\Delta) = \delta_k,$$

and in particular $\Delta \neq 0$.

Therefore $c(\omega) \neq c(\omega')$, completing the proof. □

This implies distinct division words induce distinct affine constraints.

4.3 Stepwise Realizability

Theorem 4.3 (Stepwise Realizability of Division Words)**.** *Let $\omega = (d_0, \ldots, d_{m-1})$ be a division word, and define $c(\omega)$ by*

$$c(\omega) = \sum_{j=0}^{m-1} 3^{m-1-j} \cdot 2^{\sum_{i=0}^{j-1} d_i}.$$

Then the identity

$$2^{D(\omega)} x_m = 3^m x_0 + c(\omega)$$

holds if and only if the sequence $(x_0, \ldots, x_m)$ *satisfies the accelerated Collatz relations*

$$x_{i+1} = \frac{3x_i + 1}{2^{d_i}}, \quad 0 \le i < m,$$

with each intermediate step integral.

In particular, the congruence

$$3^m x_0 + c(\omega) \equiv 0 \pmod{2^{D(\omega)}}$$

is equivalent to the existence of a trajectory realizing ω*.*

In particular, the congruence ensures that each intermediate iterate x_i is an integer, so realizability is enforced at every step.

Remark (Structural Interpretation). Each term in $c(\omega)$ corresponds to the additive contribution of a single "+1" in the Collatz iteration, propagated forward by subsequent multiplications by 3 and backward-scaled by preceding divisions by 2.

4.4 Admissibility of Division Words

Definition 4.1 (Admissibility)**.** A division word ω is admissible if the congruence

$$3^m x + c(\omega) \equiv 0 \pmod{2^{D(\omega)}}$$

has a solution.

Remark. Admissibility is therefore an arithmetic condition on the word ω itself, expressed as a divisibility constraint on the associated affine map. When this condition holds, the set of all integers x for which $F_\omega(x)$ is integral defines a unique congruence constraint modulo $2^{D(\omega)}$, corresponding to the residue class induced by ω.

4.5 Minimal Division Count

Lemma 4.4 (Minimal Division Count)**.** *For any admissible division word* $\omega = (d_0, \ldots, d_{m-1})$ *of length* m*,*

$$d_0 + \cdots + d_{m-1} \;\ge\; A020914(m),$$

where

$$A020914(m) = \lceil m \log_2(3) \rceil,$$

the minimal integer k *such that* $2^k > 3^m$*.*

Equality holds if and only if ω *achieves the minimal possible total division count among admissible words of length* m*.*

4.6 Convergent Division Word

Definition 4.2 (Convergent Division Word)**.** An admissible division word $\omega = (d_0, \dots, d_{m-1})$ is said to be *convergent* (in the linear sense) if

$$\frac{3^m}{2^{d_0+\cdots+d_{m-1}}} < 1.$$

5 Symbolic Growth and Spectral Bounds

The combinatorial complexity of admissible division words governs the number of affine constraint fibers at each depth. This growth is encoded by the sequence A186009, which counts convergent division words and hence, by injectivity of the affine encoding, the associated admissible residue classes by stopping time.

5.1 Combinatorial growth model

The sequence A186009 arises from Pascal type horizontal-sum recurrence, augmented by boundary duplication events governed by the increment grammar A022921. This grammar induces a finite symbolic system on two local operations: single increments (s) and double increments (d), whose admissible concatenations generate all configurations counted by A186009.

The resulting dynamics decompose into a finite set of admissible local patterns, whose concatenations generate all admissible configurations. The key structural feature of this grammar is that locally expansive configurations cannot occur arbitrarily often: they are separated by mandatory contractive patterns.

As a consequence, the total number of admissible configurations exhibits *submultiplicative growth*, and admits a well-defined exponential growth rate in the sense of a limsup.

5.2 Subcritical growth bound

Theorem 5.1 (Subcritical symbolic growth)**.** *Let $\widetilde{A}(i) = A186009(i+1)$ denote the number of admissible configurations of stopping time i (i.e., convergent admissible division words of length i). Then there exists a constant $\rho < 3$ such that*

$$\widetilde{A}(i) \le C\rho^i.$$

In particular,

$$\limsup_{i\to\infty} \widetilde{A}(i)^{1/i} \le \rho < 3.$$

Remark. The constant ρ arises from the maximal spectral growth of admissible cycles in the increment grammar. A complete derivation is given in Appendix A.3.

5.3 Dyadic comparison

Let $B(i) = A020914(i) = \lceil i \log_2 3 \rceil$. Thus $2^{B(i)}$ is the minimal power of 2 exceeding 3^i, and

$$3^i < 2^{B(i)} < 2 \cdot 3^i,$$

so the dyadic scaling associated with admissible constraint fibers grows asymptotically like 3^i.

Indeed, combining $\widetilde{A}(i) \leq C\rho^i$ with $2^{B(i)} > 3^i$ gives

$$\frac{\widetilde{A}(i)}{2^{B(i)}} \leq C \left(\frac{\rho}{3} \right)^i \to 0 \quad \text{as } i \to \infty,$$

since $\rho < 3$.

Thus the exponential growth of admissible configurations is strictly dominated by the exponential growth of their associated congruence moduli.

5.4 Spectral bracketing

The symbolic system admits explicit lower and upper envelope models.

Lower envelope. The restricted grammar underlying A047749, which forbids consecutive doubling events, yields an exactly solvable comparison sequence with exponential growth rate

$$\lambda_{\text{low}} = \frac{3\sqrt{3}}{2}.$$

This construction is detailed in Appendix A.3.

Upper envelope. The full admissible grammar of A186009 admits a maximal growth rate

$$\lambda_{\text{high}} = \rho < 3,$$

arising from the strictly dominant cycle structure of the increment grammar (Appendix A.3).

These bounds yield a spectral bracketing:

$$\lambda_{\text{low}} \;\le\; \limsup_{i\to\infty} \widetilde{A}(i)^{1/i} \;\le\; \lambda_{\text{high}}.$$

Remark (Reduction to a Single Growth Parameter). All combinatorial complexity of admissible division words is absorbed into the single parameter $\rho < 3$, representing the maximal exponential growth rate of the symbolic system. Subsequent arguments depend only on this subcritical growth bound, and not on the detailed structure of the enumeration.

From this point forward, the detailed combinatorics of admissible division words enter only through the growth bound $\rho < 3$.

Interpretation

The symbolic dynamics are therefore confined to a strictly subcritical regime: although the number of admissible configurations grows exponentially, its rate is bounded away from the dyadic scaling rate 3.

This establishes a fundamental imbalance: combinatorial expansion of admissible division words grows strictly slower than the dyadic refinement imposed by their congruence constraints.

This imbalance is the mechanism driving exponential sparsification in the next section.

6 Affine Constraint Fibers

6.1 Congruence Stability of Division Words

Theorem 6.1 (Congruence Stability of Division Words). *Let ω be admissible and let r_ω satisfy*

$$r_\omega \equiv -c(\omega)\cdot(3^m)^{-1} \pmod{2^{D(\omega)}}.$$

Then for all $x \equiv r_\omega \pmod{2^{D(\omega)}}$, the first m iterations of the accelerated Collatz map realize exactly the division word ω, by Theorem 4.3.

6.2 Residue Class Representation of Division Words

Theorem 6.2 (Affine Constraint Representation)**.** *Let ω be an admissible division word of length m.*

Then there exists a unique residue $r_\omega \pmod{2^{D(\omega)}}$ such that

$$x \text{ realizes } \omega \iff x \equiv r_\omega \pmod{2^{D(\omega)}}.$$

Thus admissible division words correspond to affine congruence constraints of modulus $2^{D(\omega)}$.

Proof. From the affine representation,

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}},$$

integrality is equivalent to

$$3^m x + c(\omega) \equiv 0 \pmod{2^{D(\omega)}}.$$

Since 3^m is odd, it is invertible modulo $2^{D(\omega)}$, yielding

$$x \equiv -c(\omega) \cdot (3^m)^{-1} \pmod{2^{D(\omega)}}.$$

This determines a unique residue modulo $2^{D(\omega)}$.

Conversely, any x in this class satisfies the integrality condition. By Theorem 4.3, this is equivalent to the existence of a trajectory whose successive iterates satisfy

$$x_{i+1} = \frac{3x_i + 1}{2^{d_i}}, \quad 0 \le i < m,$$

and hence realize the division word ω.

Thus admissible division words are in one-to-one correspondence with affine congruence constraints modulo powers of two. $\square$

7 Exponential Sparsification

Lemma 7.1 (Constraint Refinement). *Each admissible division word ω imposes a congruence constraint modulo $2^{D(\omega)}$, where $D(\omega) \geq A020914(m)$.*

Lemma 7.2 (Subcritical Symbolic Growth). *The number of admissible division words of length m satisfies*

$$\#\{\omega : |\omega| = m\} = O(\rho^m) \quad \textit{for some } \rho < 3.$$

Proof. This follows from Theorem 5.1, which bounds the growth of admissible configurations generated by the increment grammar. $\square$

7.1 Cylinder Structure and Prefix Compatibility

Lemma 7.3 (Injectivity of the Affine Encoding at Fixed Length). *Let $\omega = (d_0, \ldots, d_{m-1})$ and $\omega' = (d'_0, \ldots, d'_{m-1})$ be admissible division words of the same length m. Let*

$$c(\omega) = \sum_{j=0}^{m-1} 3^{m-1-j} \cdot 2^{D_j}, \quad D_j = \sum_{i=0}^{j-1} d_i,$$

and define $c(\omega')$ analogously.

If $\omega \neq \omega'$, then

$$c(\omega) \neq c(\omega').$$

Equivalently, the map $\omega \mapsto c(\omega)$ is injective on admissible division words of fixed length m.

Proof. Suppose $\omega \neq \omega'$, and let

$$k = \min\{\, j : d_j \neq d'_j \,\}$$

be the first index at which the two words differ. Then for all $j < k$, we have $D_j = D'_j$, while $D_k \neq D'_k$.

Consider the difference

$$\Delta := c(\omega) - c(\omega') = \sum_{j=0}^{m-1} 3^{m-1-j} \big(2^{D_j} - 2^{D'_j}\big).$$

The terms with $j < k$ cancel, so

$$\Delta = 3^{m-1-k}\big(2^{D_k} - 2^{D'_k}\big) + \sum_{j=k+1}^{m-1} 3^{m-1-j}\big(2^{D_j} - 2^{D'_j}\big).$$

Let $\delta_k = \min(D_k, D'_k)$. Then

$$2^{D_k} - 2^{D'_k} = 2^{\delta_k}(2^a - 2^b),$$

where one of a, b is zero and $2^a - 2^b$ is odd. Hence

$$v_2\Big(3^{m-1-k}\big(2^{D_k} - 2^{D'_k}\big)\Big) = \delta_k,$$

since 3^{m-1-k} is odd.

For $j > k$, both D_j and D'_j strictly exceed δ_k, since the prefix sums increase by positive integers. Therefore

$$v_2\Big(3^{m-1-j}\big(2^{D_j} - 2^{D'_j}\big)\Big) > \delta_k \quad \text{for all } j > k.$$

It follows that Δ is a sum of one term with 2-adic valuation exactly δ_k and remaining terms of strictly higher 2-adic valuation. Hence

$$v_2(\Delta) = \delta_k,$$

and in particular $\Delta \neq 0$.

Therefore $c(\omega) \neq c(\omega')$, completing the proof. $\square$

Lemma 7.4 (Prefix Uniqueness of Realizations)**.** *Let ω and ω' be admissible division words. If an integer x lies in both cylinders C_ω and $C_{\omega'}$, then one of the words is a prefix of the other.*

Proof. An integer x determines a unique sequence of valuations under the accelerated map f, and hence a unique division word describing its trajectory.

If $x \in C_\omega \cap C_{\omega'}$, then both ω and ω' must agree with the initial segment of this valuation sequence. Therefore, one must be a prefix of the other. $\square$

Lemma 7.5 (Prefix Descent Sufficiency)**.** *If ω is a descent word and η is any extension of ω (i.e. $\eta = \omega \star \tau$ for some nonempty suffix τ), then the cylinder C_η is a subset of C_ω. Consequently, all integers realizing η have already descended at the prefix stage ω.*

Here, $|\omega| = m$ counts the number of odd steps in the accelerated map f.

Proof. Any integer x realizing η realizes ω during its first $|\omega| = m$ accelerated iterations. Thus $x \in C_\omega$, so $C_\eta \subseteq C_\omega$. Since ω is a descent word, $F_\omega(x) < x$, and descent occurs before the suffix τ is applied. $\square$

7.2 Sparsification

Theorem 7.6 (Exponential Sparsification)**.** *The set of integers admitting a division word of length m occupies at most a proportion on the order of*

$$\frac{\#\{\textit{admissible words of length } m\}}{2^{A020914(m)}}.$$

In particular, this proportion decays exponentially in m.

By Lemma 7.4, overlaps between cylinders can only occur along prefix chains. In particular, cylinders corresponding to distinct words of the same length are pairwise disjoint

Lemma 7.7 (Level-m Coverage Bound)**.** *Let $\mathcal{W}_m$ denote the set of admissible division words of length m, and let*

$$U_m := \bigcup_{\omega \in \mathcal{W}_m} C_\omega$$

be the set of integers realizing a division word of length m.

Then

$$\mu(U_m) \;\leq\; C \left(\frac{\rho}{3}\right)^m.$$

Proof. Each word $\omega \in \mathcal{W}_m$ defines a residue class of density $2^{-D(\omega)}$, with $D(\omega) \geq B(m)$.

By Lemma 7.4, distinct words of the same length cannot be prefixes of one another, and hence their corresponding cylinders are pairwise disjoint.

Therefore,

$$\mu(U_m) = \sum_{\omega \in \mathcal{W}_m} 2^{-D(\omega)} \;\leq\; |\mathcal{W}_m| \cdot 2^{-B(m)}.$$

Using $|\mathcal{W}_m| \leq C\rho^m$ and $2^{B(m)} > 3^m$, we obtain

$$\mu(U_m) \;\leq\; C\left(\frac{\rho}{3}\right)^m,$$

as claimed. □

Lemma 7.8 (Tail Set Representation)**.** *Let E_m denote the set of integers that do not admit a descent prefix of length at most m. Then*

$$E_m \subseteq \bigcap_{k=1}^{m} U_k^c.$$

8 Density-One Descent

Lemma 8.1 (Exhaustion by Descent Prefixes)**.** *Let S_m denote the set of integers that admit no convergent division word of length $\leq m$. Then*

$$S_{m+1} \subseteq S_m, \quad \textit{and} \quad \bigcap_{m=1}^{\infty} S_m$$

is precisely the set of integers that admit no finite descent prefix.

Theorem 8.2 (Density-One Descent)**.** *The set of integers that do not admit any finite convergent division word has natural density zero.*

Proof. For each m, let $\mathcal{W}_m$ denote the set of convergent division words of length exactly m. Each $\omega \in \mathcal{W}_m$ determines a residue class modulo $2^{D(\omega)}$, hence a set of natural density $2^{-D(\omega)}$.

Let $\mathcal{U}_m$ be the union of all such residue classes:

$$\mathcal{U}_m := \bigcup_{\omega \in \mathcal{W}_m} \mathcal{C}_\omega.$$

Since distinct words of the same length correspond to distinct residue classes modulo $2^{D(\omega)}$, these classes are disjoint up to natural density, and therefore

$$\mu(\mathcal{U}_m) = \sum_{\omega \in \mathcal{W}_m} 2^{-D(\omega)}.$$

Using $D(\omega) \geq B(m)$ and $|\mathcal{W}_m| \leq C\rho^m$, we obtain

$$\mu(\mathcal{U}_m) \leq C\rho^m \cdot 2^{-B(m)} \leq C\left(\frac{\rho}{3}\right)^m.$$

Let

$$\mathcal{U} := \bigcup_{m\geq 1} \mathcal{U}_m.$$

Then

$$\mu(\mathcal{U}_m) \to 0 \quad \text{as } m \to \infty.$$

Hence the set of integers that lie in infinitely many admissible cylinders has natural density zero. Equivalently, the set of integers that avoid all finite convergent division words has density zero.

□

Corollary 8.3. *The set of integers whose trajectory under f eventually decreases has natural density one.*

Remark. Density emerges as a consequence of affine growth imbalance.

8.1 Hypothetical Zero-Density Trajectories

Any trajectory avoiding all convergent division words must lie outside the union of convergent constraint classes whose cumulative density approaches one. Such trajectories, if they exist, are therefore confined to a set of natural density zero.

In particular, no exceptional behavior can arise from any scalable arithmetic structure generated by the symbolic dynamics. Any such exception would be dynamically isolated within a sparse residual set, rather than supported by a hierarchy of positive-density constraint classes.

9 Discussion and Outlook

The preceding results establish a density-theoretic dominance of dyadic constraint refinement over the combinatorial growth of admissible division words.

Several aspects lie outside the present framework, including the fine structure of atypical trajectories within the zero-density residual set and possible refinements of the increment grammar toward sharper spectral bounds.

The affine–symbolic framework developed here is compatible with more refined local analyses, particularly in regimes where the minimal valuation structure becomes rigid.

A Division Words and Spectral Growth

This appendix collects the structural framework underlying the division-word encoding, its affine cylinder representation, the admissibility grammar governed by A020914, and the resulting growth theory for A186009. The presentation is organized in three layers: encoding, grammar, and growth.

A.1 Affine Encoding of Division Words

A division word of length $m \geq 1$ is a tuple

$$\omega = (d_0, d_1, \ldots, d_{m-1}), \qquad d_i \in \mathbb{N},$$

with total exponent

$$D(\omega) = \sum_{i=0}^{m-1} d_i.$$

Each word induces an affine transformation

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}}.$$

The constant $c(\omega)$ is defined modulo $2^{D(\omega)}$ by

$$3^m x + c(\omega) \equiv 0 \pmod{2^{D(\omega)}}, \qquad 0 \leq c(\omega) < 2^{D(\omega)}.$$

Let

$$S_j = \sum_{i=0}^{j-1} d_i, \quad S_0 = 0.$$

Then

$$c(\omega) = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{S_j}.$$

Since 3^m is invertible modulo $2^{D(\omega)}$, admissible inputs satisfy

$$x \equiv -c(\omega)(3^m)^{-1} \pmod{2^{D(\omega)}}.$$

Thus each word defines a cylinder

$$\mathcal{C}_\omega = \{r_\omega + k2^{D(\omega)} : k \geq 0\}.$$

A.2 Grammar Structure and Enumeration

Let

$$H(i) = \sum_{j=0}^{i} d_j, \qquad A(i) = \text{A020914}(i).$$

A division word is admissible iff

$$H(i) < A(i+1) \quad (i < m-1), \qquad H(m-1) = A(m).$$

Admissible words are generated by bounded backtracking over

$$d_i \in \mathbb{N}_{\geq 1}$$

subject to the prefix constraints.

For $m = 5$, with $A(1{:}5) = (2, 4, 5, 7, 8)$, there are 7 admissible words, matching A186009(6) = 7.

Each word corresponds bijectively to a cylinder:

$$\text{division word} \leftrightarrow \text{cylinder} \leftrightarrow \text{affine progression}.$$

A.3 Horizontal-Sum Dynamics and Growth

The sequence A186009 arises from

$$a_i(n) = a_i(n-1) + a_{i-1}(n),$$

with boundary modifications governed by A022921.

This induces local patterns:

$$(s, d), \quad (d, d), \quad (s, d, s), \quad (d, d, s).$$

The multiplicative factors associated with each admissible local pattern are:

Pattern	Factor
(s, d)	3
(d, d)	123/33
(s, d, s)	341/131
(d, d, s)	329/123

By monotonicity of the horizontal-sum recurrence, no admissible realization of a given pattern can exceed its extremal value, so these factors provide uniform upper bounds across all admissible configurations.

These values arise from extremal realizations of the horizontal-sum dynamics; a detailed derivation is provided in Appendix B, see in particular Section B.5.

Admissible sequences decompose into cycles. The extremal 5-cycle gives

$$\rho_5 = \left(3^2 \cdot \frac{123}{33} \cdot \frac{341}{131} \cdot \frac{329}{123}\right)^{1/5} < 3.$$

Thus

$$\limsup_{n\to\infty} A(n)^{1/n} \leq \rho_5.$$

Since $2^{A020914(n)} \sim 3^n$,

$$\frac{A(n)}{2^{A020914(n)}} \to 0.$$

Lower spectral envelope (A047749)

A047749 corresponds to the restricted grammar forbidding consecutive doubling, and pointwise:

$$A047749(n) \leq A186009(n+1).$$

Closed form:

$$a(2m) = \frac{1}{2m+1}\binom{3m}{m}, \quad a(2m+1) = \frac{1}{2m+1}\binom{3m+1}{m+1}.$$

Asymptotic growth:

$$a(n) \sim C\left(\frac{3\sqrt{3}}{2}\right)^n.$$

Hence lower rate:

$$\lambda_{\min} = \frac{3\sqrt{3}}{2}.$$

Spectral sandwich

$$\left(\frac{3\sqrt{3}}{2}\right)^n \lesssim A(n) \ll 2^{A020914(n)}.$$

division words $\Rightarrow$ cylinders $\Rightarrow$ grammar $\Rightarrow$ cycles $\Rightarrow$ spectral bounds.

B Local Growth Modes and Spectral Bounds

B.1 Finite Grammar of Local Transitions

The increment structure induced by A022921 admits a finite set of admissible local configurations:

$$\mathcal{P} = \{(s, d), (d, d), (s, d, s), (d, d, s)\}.$$

Each configuration represents a maximal locally consistent valuation pattern for the accelerated map dynamics.

B.2 Induced Affine Growth Weights

Each pattern $\pi \in \mathcal{P}$ induces an affine update of the form

$$x \mapsto \frac{3^{\ell(\pi)}x + \beta_\pi}{2^{\gamma(\pi)}},$$

where $\ell(\pi)$ and $\gamma(\pi)$ depend only on the local valuation structure.

We define the associated multiplicative growth factor

$$r_\pi := \frac{3^{\ell(\pi)}}{2^{\gamma(\pi)}}.$$

B.3 Extremal Realizations

The admissible grammar admits extremal realizations of each pattern corresponding to saturating configurations of the horizontal-sum dynamics. These realizations yield the maximal admissible growth factors:

Pattern	r_π
(s,d)	3
(d,d)	123/33
(s,d,s)	341/131
(d,d,s)	329/123

These values arise from explicit finite saturations of the horizontal-sum recurrence and serve as uniform upper bounds for all admissible instances of each pattern.

B.4 Spectral Construction

Admissible infinite sequences correspond to walks in a finite weighted directed graph whose edges are labeled by the local patterns $\pi \in \mathcal{P}$ and weighted by r_π.

The global growth rate is therefore given by the spectral radius of the associated transfer operator.

Admissible cycles correspond to closed walks in this graph, and their asymptotic contribution is given by geometric means of the associated edge weights.

B.5 Extremal Cycle and Spectral Bound

The spectral bound is obtained from a finite-cycle relaxation of the increment grammar underlying A022921. In this setting, admissible sequences correspond to walks on a finite state graph whose edges carry multiplicative weights given by the local growth factors identified in Section A.3.

Among all directed cycles in this grammar graph, the 5-cycle

$$\{s,d,s,d,d\}$$

is the 5-cycle provides an extremal candidate for maximizing the geometric mean of local growth factors. Its significance is not that it dominates all admissible sequences exactly, but that it provides an *upper-envelope relaxation*: it is obtained by removing the combinatorial constraints that would otherwise prevent indefinite repetition of locally expansive patterns.

In particular, indefinite repetition of this cycle is not admissible within the full grammar, since it violates the prefix constraints encoded by A020914. However, replacing the constrained system by this unconstrained cycle can only increase growth, and therefore yields a valid upper bound on the spectral radius.

This gives the relaxed spectral estimate

$$\rho := \left(3^2 \cdot \frac{123}{33} \cdot \frac{341}{131} \cdot \frac{329}{123}\right)^{1/5} \approx 2.976 < 3,$$

which bounds from above the true growth rate of admissible sequences.

B.6 Conclusion

All admissible growth in the symbolic system is governed by a finite weighted grammar whose spectral radius satisfies

$$\rho < 3.$$

This establishes a strict subcritical regime for symbolic expansion.

References

[1] Lothar Collatz. On the iteration of numerical functions. *Mathematische Zeitschrift*, 1937. In German.

[2] Jeffrey C. Lagarias. *The Ultimate Challenge: The 3x+1 Problem*, volume 55 of *Student Mathematical Library*. American Mathematical Society, 2010.

[3] Richard Terras. A stopping time problem on the positive integers. *Acta Arithmetica*, 30:241–252, 1976.

[4] Tomás Oliveira e Silva. Maximum excursion and stopping time record-holders for the 3x+1 problem. *Mathematics of Computation*, 68(225):371–384, 1999.

[5] Terence Tao. Almost all collatz orbits attain almost bounded values. *Forum of Mathematics, Pi*, 8:e14, 2020. doi: 10.1017/fmp.2020.7.